

GROUP A – KICK OFF MEETING DISCUSSION

Intelligent Agents Team Project

1. Simple Breakdown of assignment

Our assignment is to design a multi agent AI research assistant powered by a Large Language Model (LLM) such as ChatGPT.

We are not expected to build a fully working enterprise system. The goal is to produce a professional academic design proposal explaining:

- how the system would work
- what agents we would use
- what technologies we would use
- the challenges and risks
- why we made certain design decisions

The system should be able to:

- receive a research request
- break it into smaller tasks
- search for information
- process and summarise results
- rank useful sources
- save or present the final output

2. Proposed System Idea — “ResearchMate”

ResearchMate is a multi-agent academic research assistant that helps students and researchers gather academic information efficiently.

Example User Request:

"What are the current uses of generative AI in medical education?"

The system would then:

Step	Action
1	Break the topic into smaller research tasks
2	Search academic sources such as Semantic Scholar, Crossref, arXiv, and web search APIs
3	Filter results by relevance and credibility
4	Summarise useful papers
5	Rank sources by usefulness
6	Save/export results as CSV, JSON, Markdown, or PDF

3. Core Multi Agent Architecture

Agent	Purpose
Planning Agent	Breaks complex research goals into manageable tasks
Retrieval/Search Agent	Searches academic databases and APIs
Processing Agent	Filters, summarises, and organises findings
Ranking Agent	Evaluates relevance and credibility
Storage/Reporting Agent	Saves structured outputs
Presentation Agent	Displays final results to users

Using specialised agents improves scalability, modularity, task separation, workflow efficiency, and decision quality.

4. Roles for 4 People + Deliverables

Person	Role	Responsibilities	Deliverables
Frank	Team Lead / Project Coordinator	Coordinate team, organise meetings, manage deadlines, combine report, proofreading	Requirements section, final report, submission checklist
Alix	Research Lead / Agent Architecture	Research intelligent agents, define architecture, provide references	Architecture section, references list, diagrams
Mosleh	Technical Implementation & Workflow Design	Identify APIs/tools, explain technical feasibility, create workflow diagrams	Workflow section, technical stack, sequence diagrams
Hamad	Technical & Challenges Lead	Identify risks/challenges, explain mitigations, support evaluation	Challenges section, evaluation section, mitigation discussion

5. Proposed Workflow

- User enters research topic
- Planning Agent breaks topic into tasks
- Retrieval Agent searches for sources
- Processing Agent filters and summarises
- Ranking Agent evaluates usefulness
- Storage Agent saves the output
- Presentation Agent displays results

6. Technical Stack (Proposed)

Area	Possible Technology
LLM	ChatGPT / OpenAI API
Agent Framework	LangChain / CrewAI
Academic Search	Semantic Scholar API / Crossref / arXiv

Storage	JSON / CSV / Vector Database
Output	Markdown / PDF
Workflow	Python orchestration
Interface	Simple web dashboard

7. Required Diagrams

Diagram	Description
High-Level Architecture Diagram	Shows User, Planning Agent, Retrieval Agent, Processing Agent, Ranking Agent, Storage Agent, and Presentation Layer
Workflow / Sequence Diagram	Shows Input, Task Decomposition, Retrieval, Filtering, Summarisation, Ranking, and Output Generation

8. Best Meeting Cadence

Topic	Recommendation
Meeting Frequency	1 meeting per week
Communication	Ad hoc messages in Group A discussion forum
Meeting Duration	45–60 mins maximum
Preferred Timing	Tuesdays at 6 PM BST

9. Deliverable Structure

Meeting	Purpose	Key Deliverables
Meeting 1 (Week 3)	Kickoff & role allocation	Roles assigned, communication agreed, timelines agreed
Meeting 2 (Week 4)	Research & architecture review	Draft diagrams, references, architecture decisions
Meeting 3 (Week 5)	Final integration & review	Full draft, proofreading, Turnitin review
Week 6	Final approval & submission	Final reviewed submission

10. Final Submission Goals

- stay within 1,000 words
- demonstrate understanding of intelligent agents
- explain design decisions clearly
- include academic references
- include at least two diagrams
- demonstrate critical analysis
- explain feasibility, risks, and limitations clearly